

Trivy Use Cases

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1. Container Image Scanning

Purpose: Identify vulnerabilities in container images.

Example:

```
trivy image myapp:latest
```

2. File System Scanning

Purpose: Detect vulnerabilities in files and dependencies in local directories.

Example:

```
trivy fs /path/to/project
```

3. Git Repository Scanning

Purpose: Scan code repositories directly for vulnerabilities in dependencies or configuration files.

Example:

```
trivy repo https://github.com/example/repo.git
```

4. Kubernetes Cluster Scanning

Purpose: Check Kubernetes workloads for vulnerabilities in deployed images or configurations.

Example:

```
trivy k8s cluster
```

5. Infrastructure as Code (IaC) Scanning

Purpose: Identify misconfigurations in IaC templates such as Terraform, CloudFormation, or Kubernetes

Example:

```
trivy config /path/to/config.yaml
```

6. SBOM (Software Bill of Materials) Scanning

Purpose: Scan SBOM files to identify vulnerabilities in listed dependencies.

Example:

```
trivy sbom --file sbom.json
```

7. Runtime Vulnerability Scanning

Purpose: Monitor and scan running containers for vulnerabilities without stopping them.

Example:

```
trivy k8s cluster --live
```

8. Dependency Scanning

Purpose: Detect vulnerabilities in application dependencies (e.g., NPM, pip, Maven).

Example:

```
trivy fs /path/to/project --vuln-type library
```

9. Operating System Scanning

Purpose: Identify vulnerabilities in operating system packages.

Example:

```
trivy fs / --vuln-type os
```

10. Continuous Integration (CI) Integration

Purpose: Automate vulnerability scanning in CI/CD pipelines.

Example (GitHub Actions):

steps:

- name: Run Trivy Scan

uses: aquasecurity/trivy-action@v0.4.0

with:

image-ref: 'myapp:latest'

11. Policy Management and Enforcement

Purpose: Define and enforce security policies for deployments.

Example:

```
trivy fs /path/to/project --severity HIGH,CRITICAL
```

12. Export and Reporting

Purpose: Generate detailed vulnerability reports in various formats.

Example:

```
trivy fs /path/to/project --format html --output report.html
```

By leveraging these use cases, Trivy can be incorporated across different stages of software development.